

Rural water point siting in Mayuge District, Uganda

SUBJECT	Placement of 10 additional water points under a fixed budget
PREPARED BY	An autonomous analysis agent, from open data, without field verification
DECISIONS	All 8 decisions in this run were recorded by a person.
REVIEWED BY	Not reviewed. No planner overrides were applied to this run.
GENERATED	2026-09-04T15:59:05Z
RUN	mayuge-water-20260904T155905Z
PROVENANCE	ac15a4eae9de9c52

Statement

An estimated 570,781 people live in the area of interest. 371,949 of them, 65.2%, live within about 21 minutes of walking over the modelled terrain of a water point recorded as serving. 198,832 people, 34.8% of the district, do not (Ex01, Ex02, Ex03).

Of the 1,251 water point records held for this district, 354 are recorded as serving and 897 are not (Ex01). The median record is 1.9 years old, so the figures above describe the surveyed state of the network rather than its state today (Ex10).

2,095 candidate locations were evaluated (Ex04). Selecting 10 of them by maximum coverage brings a further 60,208 people inside the service radius, raising coverage to 75.7% (Ex05, Ex06). Returns fall away quickly: the first site reaches 12,365 people and the last 4,207, so the number of sites to build is a decision that should be taken on the curve at Ex05 rather than on a round number.

No planner overrides were applied. The recommendation at Ex06 is the unmodified output of the selection method at Ex05.

4 of the 10 independent checks at Ex08 returned a flag rather than a pass: data currency, boundary effect, equity, cartographic consistency. The flagged findings are reproduced in full at Ex08 and are not resolved in this bundle.

3 properties of the source registers would have changed a coverage figure had they been handled differently. Each is recorded at Ex10 with what was observed and what was done about it. None were corrected silently.

Recommendations in this bundle are advisory. Land availability, tenure, water table, community consent and construction feasibility are outside the data used here and inside the judgement of the officer reviewing it. No figure in this bundle is asserted without an exhibit number after it.

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Verification

Every retrieval in this bundle is recorded with its endpoint, the query as executed, the time of retrieval and the record counts before and after cleaning (Ex01, Ex02, Ex11). Cleaning rules are declared per source in the handbook files that accompany this bundle and are applied in order, each reporting its own drop count, so no record leaves the pipeline unaccounted for.

Coverage is recomputed independently of the solver by a separate process before this bundle is produced, and that process can prevent it being produced at all (Ex08). Coverage is always the union over serving locations; a per-location sum double counts every settlement reachable from two of them, and Ex08 reports what that sum would have claimed.

Properties of the source data that would change a figure if read differently are recorded at Ex10 rather than handled silently. Where the agent's own source handbook was found to be incomplete, that is recorded there too.

No figure in this bundle was estimated, interpolated or supplied from memory. A figure is either computed from a recorded retrieval or it is absent.

Ex00 – Decisions taken, and by whom

SUPPORTS	That the judgements this assessment rests on are attributable
CAPTURED	Recorded before the analysis ran
METHOD	Each row is a judgement about values rather than a technical parameter: what counts as served, whose need the programme answers, whether a register of this age is fit to direct spending. The analysis refuses to run without them rather than assuming one, because assuming one would move the decision away from the officer accountable for it. Where a row is attributed to the agent, no district position was recorded and the agent decided in its absence.

All 8 decisions in this run were recorded by a person.

Decision	Value	Recorded by	Reason given
service_radius_m	1000.0 metres	Jadey	1000 m Mayuge already reports against,
coverage_basis	walking_time	Jadey	walking time : Mayuge walk around water rather than across it
objective	max_coverage	Jadey	198,832 unserved is too many to spend a small programme on the margins
budget	10 facilities	Jadey	we have varied budget, we want to find an ideal number
data_currency_accepted	yes	Jadey	conditional, some will have been repaired and some of the 354 will have failed
coverage_tolerance	0.05 share of achievable coverage	Jadey	198,832 unserved is too many to spend a small programme on the margins
review_floor	6.5 weighted score out of 10	Jadey	don't be too strict
equity_accepted	unresolved	not settled at run time	The measured distribution was reported to the operator and no position was recorded, so the assessment states it as unresolved rather than accepted.

Ex01 – Rural water point siting register for Mayuge

SUPPORTS	The installed stock and its recorded condition
CAPTURED	https://data.waterpointdata.org/resource/eqje-vguj.json on 2026-09-04, read from the local cache for this run
METHOD	Socrata query below, executed verbatim. Cleaning rules are declared in handbooks/wpdx.yaml and applied in order; each rule reports its own drop count.

```
clean_country_name = 'Uganda' AND clean_adm2 = 'Mayuge'
```

Records returned	1,378
Records after cleaning	1,251
Discarded	127

RECORDS DISCARDED, BY RULE

flagged duplicate by WPdx	127
Recorded as serving	354
Recorded as not serving	897
Median record age	1.9 years

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Ex02 – Population surface

SUPPORTS	The denominator for every coverage figure in this bundle
CAPTURED	https://hub.worldpop.org/rest/data/pop/wpgp on 2026-09-03, read from the local cache for this run
METHOD	National 100 m grid, read over the area of interest and aggregated into blocks. The aggregation is coarser than the source and finer than the service radius, so it does not bias coverage.

UGA ppp 2020, bbox 33.2835 -0.0474 33.7104 0.6061 (WGS 84), aggregated 5x5

Raster cells in window	15,912
Aggregated demand cells	8,992
Population in the area of interest	570,781

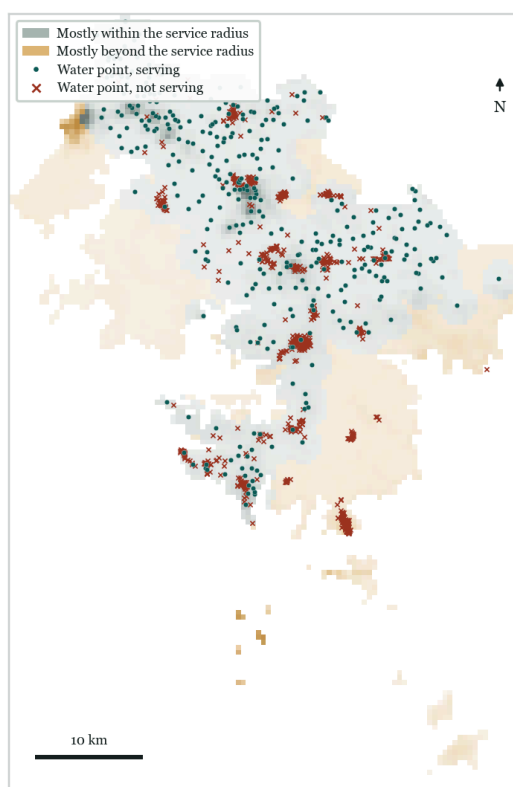
population year 2020; 401,408 source cells at ~100 m read over the window, aggregated 5x5 into 15,912 cells of ~500 m

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Ex03 – Coverage before any intervention

SUPPORTS	The size of the gap the recommendation is trying to close
CAPTURED	Computed from Ex01 and Ex02
METHOD	A demand cell counts as covered when it lies within about 21 minutes of walking over the modelled terrain of a point recorded as serving. Coverage is the union over serving points; a per-point sum would double count every settlement reachable from two of them.

Population	570,781
Covered today	371,949 (65.2%)
Not covered	198,832 (34.8%)



Serving and non-serving points over the population surface. Warm shading is population beyond the service radius.

Ex04 – Candidate sites considered

SUPPORTS	That the recommendation was selected from a stated, reproducible set
CAPTURED	Constructed from Ex01 and the area of interest
METHOD	Two kinds of candidate: an existing point recorded as not serving, which could be rehabilitated, and a regular lattice of greenfield locations. Lattice points already inside an existing service area are removed before selection.

Candidates evaluated	2,095
Lattice spacing	750 m
Minimum spacing between recommendations	1000 m

Ex05 – Selection method and its guarantee

SUPPORTS	That the ranking is reproducible and its optimality is bounded
CAPTURED	Computed
METHOD	Candidates are scored by the population they would newly bring inside the service radius and selected greedily. On a monotone submodular coverage function greedy selection is within a factor of $1 - 1/e$, about 63 per cent, of the optimum. The exact integer programme is not used in the loop: it takes minutes at district scale where the heuristic takes seconds, and every planner override triggers a re-solve.

Sites selected	10
Population newly covered	60,208
Coverage after the programme	75.7%
Benchmark against an exact solver	not run (not run)

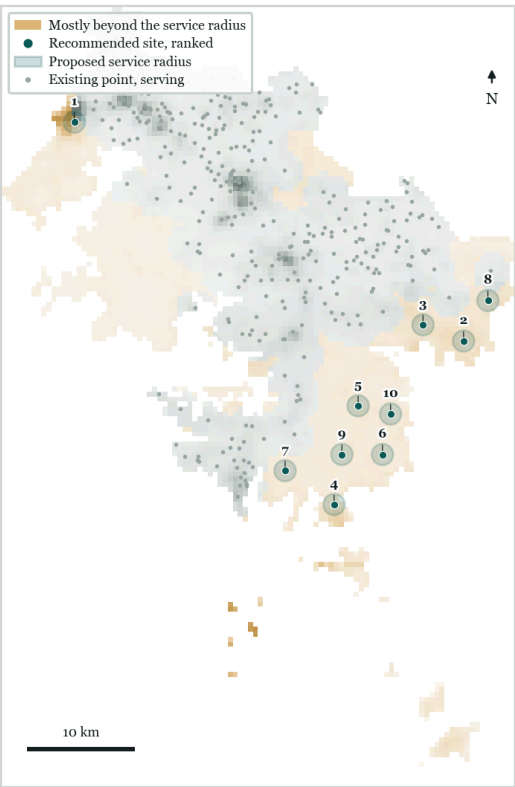
Sites built	Newly covered	Cumulative	Share of district
1	12,365	384,314	67.3%
2	7,240	391,554	68.6%
3	6,715	398,268	69.8%
4	6,100	404,368	70.8%
5	5,076	409,444	71.7%
6	4,775	414,219	72.6%
7	4,609	418,828	73.4%
8	4,589	423,417	74.2%
9	4,533	427,950	75.0%
10	4,207	432,157	75.7%

Marginal and cumulative population covered. The first site reaches 12,365 people, the last 4,207.

Ex06 – Recommended sites

SUPPORTS	The recommendation itself, in coordinates
CAPTURED	Computed
METHOD	Sites in selection order. Each row's coverage is additional to every row above it, so the programme must be built from the top down. Coordinates are WGS 84.

No.	Site	Type	Location (WGS 84)	Newly covered	Nearest serving
1	S-001	greenfield	0.5090, 33.3442	12,365	2.1 km
2	S-002	greenfield	0.3258, 33.6677	7,240	4.3 km
3	S-003	greenfield	0.3393, 33.6340	6,715	3.5 km
4	S-004	rehabilitate	0.1889, 33.5601	6,100	7.5 km
5	S-005	greenfield	0.2715, 33.5800	5,076	5.4 km
6	S-006	greenfield	0.2308, 33.6003	4,775	8.4 km
7	S-007	greenfield	0.2172, 33.5194	4,609	3.0 km
8	S-008	greenfield	0.3597, 33.6879	4,589	2.8 km
9	S-009	greenfield	0.2308, 33.5666	4,533	4.8 km
10	S-010	greenfield	0.2647, 33.6070	4,207	8.5 km



Recommended sites and their service radii over the unserved population.

BASIS FOR EACH SITE

S-001	New site covering 12,365 people currently beyond the service radius; nearest working point 2.1 km away.
S-002	New site covering 7,240 people currently beyond the service radius; nearest working point 4.3 km away.
S-003	New site covering 6,715 people currently beyond the service radius; nearest working point 3.5 km away.
S-004	Rehabilitate the existing point covering 6,100 people currently beyond the service radius; recorded as non-functional; technology on record: Public Tapstand; nearest working point 7.5 km away.
S-005	New site covering 5,076 people currently beyond the service radius; nearest working point 5.4 km away.
S-006	New site covering 4,775 people currently beyond the service radius; nearest working point 8.4 km away.
S-007	New site covering 4,609 people currently beyond the service radius; nearest working point 3.0 km away.
S-008	New site covering 4,589 people currently beyond the service radius; nearest working point 2.8 km away.
S-009	New site covering 4,533 people currently beyond the service radius; nearest working point 4.8 km away.
S-010	New site covering 4,207 people currently beyond the service radius; nearest working point 8.5 km away.

Ex07 – Planner overrides and what they cost

SUPPORTS CAPTURED METHOD	That the recommendation was reviewed by a person, and where it was overruled overrides YAML, reproduced verbatim below Each override was applied on its own and the plan re-solved, so every decision carries its own cost rather than one lumped figure. Overrides are never blocked. A veto removes an area, not a single lattice point, because a planner saying not there means a place.
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No overrides were applied to this run. The recommendation in Ex06 is the unmodified output of Ex05.

Ex07b – Safety gate decisions

SUPPORTS CAPTURED	That access, transfer and fallback decisions were taken deliberately Recorded at the point of each decision
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METHOD A gate never silently permits. Each row is a point at which the run either proceeded and recorded why that was safe, or refused. A refusal removes a source from the analysis and is stated here rather than left for the reader to infer from a missing figure.

Gate	Outcome	Decided by	Reason recorded
source authorisation	PASS	harness	Water Point Data Exchange Plus (WPdx+) is a public endpoint; no authorisation required
source authorisation	PASS	harness	WorldPop gridded population is a public endpoint; no authorisation required
guardrail breach	PASS	harness	Overrides forgo 0 people, 0.0% of the achievable coverage, within the 5% tolerance set for this run.

Ex08 – Independent checks

SUPPORTS CAPTURED METHOD That the plan was checked by a process that did not produce it
 Computed
 These checks run in a separate process from the one that produced the plan, can reject it, and do not read the override reasons in Ex07. They check the plan against the data, not against the account given of it. A single rejection stops the bundle being produced.

Check	Result	Finding
coverage arithmetic	PASS	union recount matches at 432,157; across the 10 new sites a per-facility sum would report 68,916 against a true union of 68,602, an overcount of 313
geometry	PASS	all 10 sites lie inside the area of interest
budget	PASS	10 sites within the budget of 10
coordinate reference system	PASS	EPSG:32636 (WGS 84 / UTM zone 36N); over 400 sampled pairs the projected distance departs from great-circle by 0.317% at the median and 0.595% at most, or about 6 m on a 1000 m service radius
data currency	FLAG	median source record is 1.9 years old. The age was put to the accountable officer and accepted: conditional, some will have been repaired and some of the 354 will have failed
aggregation sensitivity	PASS	reweighting the demand surface within cells moves the coverage claim from 75.7% to 75.7%, a shift of 0.02%
boundary effect	FLAG	2 of 10 sites lie within one service radius of the district boundary; the nearest is 625 m from it. Population and facilities across the boundary are not represented, so coverage at these sites may be misstated
equity	FLAG	59% of newly covered people live in the densest quartile of cells, against 42% district-wide; the plan favours dense settlements over remote ones
provenance	PASS	4 sources recorded, all figures traceable
cartographic consistency	FLAG	the rendered figures were not reviewed against the account; run the map-reviewer agent over figures.json and pass its verdict with --figure-review

Ex08b – Scoring review

SUPPORTS CAPTURED METHOD That the account was scored against a stated floor before issue
 Computed by a reviewer that reads only this bundle's results file
 The reviewer receives the results file and nothing else: not the solver, not the code path, not the override reasons. It cannot score the intention behind the plan, only the account given of it. Every dimension carries a floor and the weighted average carries a floor; the run is issued only when all are met, or when a person forces it and that is recorded.

Dimension	Score	Floor	Weight		Basis
data adequacy	8.5	5.0	0.25	met	median record 1.9 years old, put to the accountable officer and accepted with a recorded reason; 1 semantic anomalies in the register
method fitness	7.5	5.0	0.20	met	On a monotone submodular coverage function greedy selection is within a factor of $1 - 1/e$, about 63 per cent, of the optimum.; no exact benchmark (not run); reach measured as walking time from the local friction value at each candidate, not as a least-cost traverse
spatial rigour	7.0	6.0	0.25	met	6 of 6 required diagnostics present; distances in EPSG:32636 (WGS 84 / UTM zone 36N)
accountability	9.0	6.0	0.20	met	every figure resolves to a recorded retrieval; 3 source anomalies disclosed
actionability	9.5	4.0	0.10	met	10 sites with coordinates and a stated basis; marginal coverage curve supports a budget decision; 3 sensitivity scenarios

Weighted score	8.12
Floor to issue	6.50
Decision	issue
Reviewer	deterministic rules over results.json

Ex09 – Sensitivity of the recommendation

SUPPORTS	How far the recommendation depends on assumptions the planner can change
CAPTURED	Computed
METHOD	Each scenario re-solves from the same data with one assumption changed. A recommendation that survives every row is one a planner can defend.

Scenario	Covered	Share	Against the base case
Rehabilitation of existing points only	398,352	69.8%	-33,805
Half the budget (5 sites)	409,444	71.7%	-22,713
Double the budget (20 sites)	469,384	82.2%	+37,227

Ex10 – Anomalies in the source data, and how they were handled

SUPPORTS	That the source registers were read rather than assumed
CAPTURED	Observed during retrieval and cleaning
METHOD	Every item below is a property of the published data that would change a coverage figure if handled differently. Each is recorded with what was observed, what the agent did, and what a reader must keep in mind. None of these were corrected silently.

1. SEMANTICS · WPdx+

Observed	status_clean is not two-valued in this district. Values present: Non-Functional (891); Functional, needs repair (354); Functional, not in use (6).
Handling	Serving status is read from the status_semantics block in handbooks/wpdx.yaml, which treats Functional and Functional, needs repair as serving and everything else as not serving.
Bearing	An equality test against the string Functional would score every point in this district as not serving and inflate the gap.

2. DUPLICATION · WPdx+

Observed	127 of 1,378 returned records were removed: flagged duplicate by WPdx.
Handling	Removed before any coverage computation, per the cleaning rules declared in the handbook.
Bearing	Coverage computed on the raw response would count these records twice or count superseded observations.

3. METHOD · MAP friction surface

Observed	Over this area the walking friction surface implies 21 minutes per kilometre at the median and 60 at the 90th percentile, so the chosen 1000 m corresponds to about 21 minutes of walking.
Handling	Coverage is tested as walking time rather than straight-line distance. Reach is computed per candidate from the local friction value rather than by a least-cost accumulation from every candidate, which would be exact but is not tractable at this candidate count.
Bearing	Where terrain changes sharply within one facility's reach, the local-friction approximation applies the conditions at the site in every direction, and is conservative rather than optimistic. The surface is modelled at about one kilometre and does not represent local footpaths.

Ex11 – Reproduction record

SUPPORTS	That this bundle can be regenerated from the recorded queries
CAPTURED	Computed
METHOD	The bundle is a function of one results file and the manifest that produced it. Any change to a source query, a cleaning rule or a retrieval date changes the provenance hash.

Run	mayuge-water-20260904T155905Z
Generated	2026-09-04T15:59:05Z
Provenance hash	ac15a4eae9de9c52

```
python -m siting.cli --country Uganda --adm2 Mayuge --iso3 UGA --domain water --mode manual --out
sessions/mayuge-20260904T152614Z-6127d5/runs --format both --reviewer rules --decisions sessions/
mayuge-20260904T152614Z-6127d5/decisions.yaml
```

The --out and --decisions paths above have been shortened to be relative to the repository root. This run was produced under sessions/mayuge-20260904T152614Z-6127d5/ and the directory was copied to sample-runs/mayuge-uganda/ for publication. Only the machine-specific path prefix was removed; the manifest hash in manifest.json was recomputed over the shortened command, and nothing else in this record was altered.

Source	Licence	Accessed
Water Point Data Exchange (WPdx+), global water point registry, https://www.waterpointdata.org/	CC BY 4.0	2026-09-04
GADM database of Global Administrative Areas, version 4.1, https://gadm.org	GADM licence, free for academic and non-commercial use	2026-09-03
WorldPop, School of Geography and Environmental Science, University of Southampton, https://www.worldpop.org	CC BY 4.0	2026-09-03
Weiss, D.J. et al. Global maps of travel time to healthcare facilities. Nature Medicine 26, 1835-1838 (2020).	CC BY 4.0	2026-09-04